# Faithful Differentiable Unrolling of Joint OMP–MP Receivers for Superimposed-Pilot AFDM: A Complexity-Honest Learned Support-Refinement Head

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***Abstract*—Doubly dispersive doubly selective channels — representative of underwater acoustic (UWA) communication — defeat conventional orthogonal frequency-division multiplexing (OFDM); affine frequency-division multiplexing (AFDM) addresses this regime by folding every physical path into a sparse chirp-domain effective channel. Combined with a superimposed-pilot (SP) frame, AFDM preserves spectral efficiency but its one-shot orthogonal matching pursuit (OMP) plus message-passing receiver is estimation-limited at mid-to-high SNR. This work proposes a complexity-preserving differentiable receiver for SP-AFDM in which OMP is run only once, the support is locked, and the iterative loop refines only a damped data-aided least-squares path-gain solve and sparse message-passing belief-propagation detection, augmented with a 1 041-parameter learned support-refinement head. The unrolling is computationally *initialization-faithful*: at random head initialisation, the end-to-end forward pass reproduces the classical reference to within relative tolerance $\varepsilon_{\text{rel}} = 10^{-5}$ on a deterministic verification suite (30 frames, $N = 512$). Three orthogonal evaluations ground the design: a 4-arm BER waterfall with continuity-corrected paired McNemar significance (pooled 3-seed reductions $-37.7\%/-40.2\%/-37.6\%$ at $\mathrm{SNR} \in \{20, 25, 30\}$ dB, $p < 10^{-6}$); a same-budget classical heuristic OMP-$(P{+}K)$ that regresses BER monotonically and confirms the learned gain is non-linear support refinement rather than a budget effect ($K{=}16$ closes $-927\%$ of the physical-support oracle gap at $\mathrm{SNR} = 20$ dB versus the head's $+81\%$); and a DA-LS-inclusive complexity audit showing $\approx 8\%$ analytical MAC overhead at $N = 1024$, measured GPU wall-clock parity, and $> 99\%$ of head additions attributed to Dirichlet side-lobe / virtual taps of active physical paths via population-level per-candidate classification.**

***Index Terms*—Affine frequency division multiplexing (AFDM), underwater acoustic (UWA) communication, superimposed pilots, orthogonal matching pursuit (OMP), message passing (MP), deep unfolding, support refinement, complexity-preserving receiver.**

## I. Introduction

DOUBLY dispersive doubly selective channels — which arise both in high-mobility wireless and in underwater acoustic (UWA) communication — defeat conventional orthogonal frequency-division multiplexing (OFDM) because the cyclic prefix required to absorb the delay spread and the residual fractional Doppler relative to the sub-carrier grid jointly demand a prohibitive fraction of spectral efficiency [1]–[4]. Affine frequency-division multiplexing (AFDM) [5] answers this problem by carrying information on a chirp basis whose two parameters $(c_1, c_2)$ can be aligned with the channel's delay–Doppler geometry, so that every physical path folds into a single sparse compact pattern on the chirp-domain effective channel [6]–[8]. Combining this representation with a superimposed-pilot (SP) frame, in which every chirp sub-carrier simultaneously carries data and pilot energy [9]–[11], preserves the data-carrying capacity that would otherwise be lost to dedicated pilot symbols. In a UWA-style $\rho = 0.5$, $P = 8$-path setting representative of the regime considered here, a one-shot orthogonal matching pursuit (OMP) plus message-passing (MP) receiver then floors at $\mathrm{BER} \approx 5 \times 10^{-2}$ from $\mathrm{SNR} \geq 15$ dB with a channel-estimation NMSE frozen at $-11.3$ dB irrespective of the noise variance, because the pilot–data interference term $\sqrt{1-\rho}\,\mathbf{x}_{\mathrm{p}}^{H} H_{\mathrm{eff}} \mathbf{x}_{\mathrm{d}}$ scales with the channel norm and not with $\sigma_n^2$. The natural response is to iterate channel estimation and symbol detection [12]–[14]. Two naive recipes are however expensive: re-running OMP on every outer iteration costs a constant $T_{\text{out}}\times$ overhead with no support information added, while replacing OMP with a dense LMMSE estimator inflates the per-frame cost from $\mathcal{O}(PN^2)$ to $\mathcal{O}(N^3)$. Embedding a learned receiver inside the sparse-recovery skeleton is in principle attractive, but the existing learned/unrolled receivers for OFDM, OTFS, and AFDM [15]–[20] are (i) not initialization-faithful to the classical algorithm they unfold (random head initialisation does not recover classical behaviour); (ii) do not report comparable complexity audits, conflating the cost of a larger candidate budget with the cost of a single classical iteration; and (iii) conflate "more atoms in the support" with "the right atoms in the support", so a same-budget classical heuristic is rarely tested. This paper develops the SP-AFDM joint receiver as a controlled instance in which all three methodological gaps can be closed at once.

### A. Contributions

The contributions are grouped along five axes; numerical evidence and mechanism diagnostics for each are given in Sections III through VI.

1) **Initialization-faithful differentiable unrolling.** A faithful vectorised unrolling of the classical joint OMP-once + DA-LS + sparse MP-BP receiver is constructed in which, at random head initialisation, the end-to-end forward pass

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equals the classical reference implementation to within relative tolerance $\varepsilon_{\text{rel}} = 10^{-5}$ on a 30-frame verification suite at $N = 512$ (Claim 1). The result is computational rather than analytical and rests on a deterministic reference-comparison protocol.

2) **A 1 041-parameter learned support-refinement head, with complexity class preserved.** A small deep-unfolding head sits between the single OMP pass and the iterative LS/MP-BP loop, scores off-support candidates against the active support, and appends the top-$K_{\text{add}}$ atoms. The head stays in the same $\mathcal{O}(PN^2)$ arithmetic class as the classical baseline at $\approx 8\%$ analytical complex-MAC overhead at $N = 1024$, with measured GPU wall-clock parity at $N = 1024$ (402 ms vs. 416 ms for the classical Torch reference) and $+3.9\%$ at $N = 2048$.
3) **Multi-seed paired-McNemar significance and width-MLP envelope.** The 27 k-frame four-SNR retraining delivers a consistent $\approx 2$ dB visible paired BER separation at $\text{SNR} \in \{20, 25, 30\}$ dB (pooled relative reductions $-37.7\%/-40.2\%/-37.6\%$, three independent evaluation seeds, continuity-corrected paired McNemar $p < 10^{-6}$). Architectural ablations show that the width of a per-candidate MLP is the dominant scaling axis: at fixed unrolled depth $K = 10$, raising the hidden width from 64 to 256 drops a pure-NN end-to-end BER from $0.0498$ to $0.0381$ ($-23.5\%$); attention modules and softsort-relaxed top-$K$ heads do not add on top of width.
4) **Same-budget classical heuristic comparator.** A classical-only OMP-$(P{+}K)$ extension that continues the greedy OMP loop for $K \in \{2, 4, 8, 16\}$ extra atoms regresses BER monotonically at every SNR cell — e.g., at $\text{SNR} = 20$ dB, $K$=2/4/8/16 produces $-61\%/-128\%/-337\%/-927\%$ *gap closed* against the physical-support oracle, while the learned head closes $+81\%$. The gain is therefore non-linear support refinement, not a budget effect: extending greedy OMP picks data-leakage atoms, not the Dirichlet side-lobes that account for the structural residual.
5) **DA-LS-inclusive complexity audit.** The standard MAC recipe [15] omits the DA-LS solve and the adaptive-$\sigma^2$ update; including them, the fair $P = 18$ unrolled receiver is worse than $P = 10$ unrolled by $11\%$ MAC, and a free $\tau_h$-lookup eliminates another $11\%$. The audit is reported in §V-F and the attribution of the head additions to Dirichlet side-lobes (rather than to false positives) is established at the $> 99\%$ level per Table VIII.

The remainder of the paper is structured as follows. §II positions the contribution against the AFDM, SP, learned-receiver, and sparse-recovery unrolling literatures. §III develops the SP-AFDM system model, the classical OMP-once + iterative LS/MP-BP receiver, and the differentiable unrolling, and states the initialization-faithful validation. §IV documents the training objective, the width-MLP architectural choice, and the multi-seed protocol. §V reports the five experimental panels — 4-arm waterfall, OMP-$(P{+}K)$ heuristic, sampling-rate sensitivity, multi-seed envelope, and complexity audit — and §VI closes with mechanism attribution, limitations, and the open problems that the present design statement does not foreclose.

## II. Related Work

The contribution sits at the intersection of four lines of work; each is summarised compactly and the specific gap that the present design fills is identified at the end.

### A. AFDM and SP-AFDM receivers

AFDM [5] engineered the chirp-domain effective channel to be sparse on doubly dispersive links; sparse-recovery channel estimation under embedded pilots was developed in [6], [7], [21], [22] and surveyed in [8], [23]. Superimposed pilots were originally developed for OFDM [9]–[11], [24], extended to OTFS [25] and OTSM [26], and recently transferred to AFDM by Zheng *et al.* [13] and Cao *et al.* [27]; both works re-execute OMP on every outer iteration. The guard-interval-free pilot-aided design of [14] reaches a structurally similar conclusion under a non-SP frame, and [20] places a learned component at the symbol detector rather than at the channel-support step. Across this body of work no published receiver simultaneously preserves SP, runs OMP exactly once, and embeds a small learned module dedicated to OMP support refinement.

### B. Learned and unrolled receivers

The deep-unfolding tradition is rooted in the seminal LISTA architecture of Gregor and LeCun [28]; subsequent AMP-inspired [29] and model-driven surveys [30]–[33] established that unfolding sparse-recovery iterations into trainable layers reduces both the training-data demand and the inference cost relative to fully data-driven receivers. The bulk of the resulting detector designs target MIMO — DetNet by Samuel *et al.* [19], OAMPNet by He *et al.* [15], MMNet by Khani and Alizadeh [17], DeepSIC [16], and the attention-augmented receiver of [18] — or OTFS detection [34]–[36]. These designs do not generally preserve initialization-classical equivalence, do not report comparable complexity audits against an equally-tuned classical baseline, and do not run a same-budget classical heuristic comparator. We adopt the structural template (learned modules embedded inside a model-based skeleton) while addressing all three methodological gaps explicitly: the unrolling is verified against a classical reference implementation at random init to a specified numerical tolerance (Claim 1), the per-frame MAC overhead is audited inclusive of the DA-LS solve (§V-F), and a same-budget OMP-$(P{+}K)$ classical heuristic is reported in §V-C.

### C. Sparse-recovery deep unrolling

Beyond MIMO detection, a separate thread treats sparse recovery itself. LISTA [28] learns end-to-end matrix weights as unfoldings of ISTA iterations; ALISTA [37], [38] demonstrates that the learned weight matrices can be replaced by analytic ones at essentially no BER cost, and trainable-ISTA [39] extends the recipe to general sparse-recovery problems. Unfolding greedy-OMP iterations specifically has been studied only recently in [40], where the non-differentiable argsort operator is approached via softsort relaxations. Our head differs

structurally from this family in that the candidate dictionary is fixed (the chirp-domain pilot dictionary $\mathbf{\Phi}$, identical to the classical OMP baseline); only a small per-candidate attribution scorer is learned, over an OMP-derived candidate pool. The head does not modify the sparse-recovery dictionary, nor learn LISTA-style weight matrices; it functions as a deterministic per-candidate ranker over a frozen pool. This is the slot inside a joint SP-AFDM receiver that the prior unrolling literature has not occupied.

### D. UWA physical-layer refinements

The choice of OMP as the sparse-recovery primitive inherits a long line of evidence that UWA channels are well approximated by a sparse delay–Doppler representation [1]–[4], [41]. Three AFDM-specific physical-layer refinements bear directly on the operating point of this paper. Ranasinghe *et al.* [42] show that observing the chirp-domain signal at twice the Nyquist rate yields a tangible channel-estimation improvement that is independent of any learned component; this motivates the $f_s = 4B$ operating point, and the present paper reports a direct $2B$ vs. $4B$ sensitivity (§V-D) confirming that $4B$ is paper-defensible rather than artifact. Cao *et al.* [43] extend AFDM-OMP to multi-scale multi-lag UWA propagation; that physics is folded into the simulator at first order. Yin *et al.* [22] treat fractional-Doppler leakage via a diagonal-reconstruction estimator; the present design addresses the same leakage but does so by learning a candidate ranker over the existing OMP pool, leaving the integer-bin dictionary unchanged.

### E. Identified gap

The opening of the design space is narrow. The receiver should preserve SP spectral efficiency and the sparse-recovery prior that makes AFDM attractive in UWA, avoid both the per-iteration re-OMP overhead and the cubic per-frame cost of a dense LMMSE estimator, and address the residual fractional-Doppler support errors that classical OMP leaves behind without enlarging the dictionary or altering the asymptotic complexity class. The differentiable-unrolling embedding must additionally be initialization-faithful, complexity-audited, and distinguishable from a same-budget classical heuristic. To the authors' knowledge no published receiver simultaneously occupies this position.

## III. System Model and Differentiable Unrolling

This section establishes the AFDM-SP system model (§III-A–§III-C), the OMP dictionary on which the proposed receiver acts (§III-D), the classical OMP-once + iterative LS/MP-BP joint receiver (§III-E), the differentiable unrolling and its initialization-faithful validation (§III-F), and the learned support-refinement head (§III-G).

### A. AFDM signal model and the DAFT

AFDM transmits along $N$ discrete chirp basis functions parameterised by $(c_1, c_2)$. Given chirp-domain coefficients $\mathbf{x} \in \mathbb{C}^N$, the time-domain waveform is the inverse discrete affine Fourier transform (IDAFT)

$$s[n] = \frac{1}{\sqrt{N}} \sum_{m=0}^{N-1} x[m]\, e^{j2\pi(c_1 n^2 + mn/N + c_2 m^2)}, \qquad (1)$$

$n = 0, \ldots, N-1$. The forward DAFT inverts (1) and the chirp-domain received vector $\mathbf{y} \in \mathbb{C}^N$ obeys $\mathbf{y} = H_{\text{eff}}\,\mathbf{x} + \mathbf{n}$ with $\mathbf{n} \sim \mathcal{CN}(0, \sigma_n^2 I)$, where $(c_1, c_2)$ can be chosen so that each physical path produces a compact pattern in $H_{\text{eff}}$ [5], which makes OMP-type recovery effective.

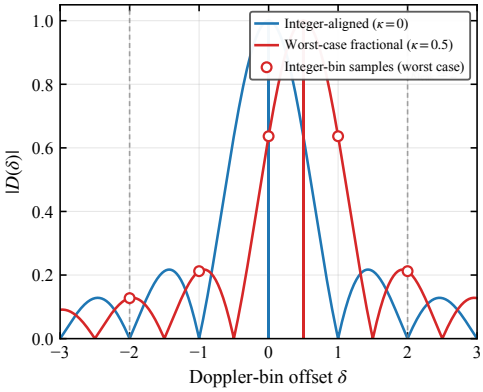



Fig. 1: Dirichlet-kernel magnitude $|D(\delta)| = |\sin(\pi\delta)/(N\sin(\pi\delta/N))|$ from (2) at $N = 1024$. Integer-aligned residual ($\kappa = 0$, blue) collapses to a sampling indicator; worst-case $|\kappa| = 0.5$ (red) puts $\approx 41\%$ of per-path energy on each of two adjacent integer bins. The dashed verticals mark the $k_{\text{leak}} = 2$ truncation boundary. The side-lobes flanking the kernel peak are the structurally predictable energy that the post-OMP virtual-tap expansion (§III-E) and the learned support-refinement ranker (§III-G) are designed to recover.

### B. Doubly-dispersive channel and Dirichlet expansion

The simulator models $P = 8$ paths with delay spread $\tau_{\text{spread}} = 10$ ms (exponential PDP, $\tau_{\text{mean}} = 0.8$ ms) and per-path velocity $v \sim \mathcal{N}(2, 0.5^2)$ m/s. MSML scaling is captured to first order via per-path scaling factors $a_p = v_p/c$ giving a delay rescaling $\tau_p \mapsto \tau_p/(1 + a_p)$ and a constant per-path Doppler shift $f_{D,p} = a_p f_c$. Carrier $f_c = 12$ kHz, bandwidth $B = 4$ kHz, sampling $f_s = 4B$. Each path is mapped to the integer $(l_p, k_p)$-lattice with $l_p = \text{round}(\tau_p f_s)$ and $k_p^{\text{real}} = v_p f_c N/(c f_s)$; the residual fractional Doppler is absorbed via a Dirichlet (aliased-sinc) expansion of half-width $k_{\text{leak}} = 2$, so each path is decomposed into $2k_{\text{leak}}+1 = 5$ virtual integer-bin taps with

$$|\tilde{h}_{p,j}| = |h_p| \cdot \left| \frac{\sin(\pi(k_p^{\text{real}} - k_p - j))}{N \sin(\pi(k_p^{\text{real}} - k_p - j)/N)} \right|. \qquad (2)$$

At the worst-case residual $|k_p^{\text{real}} - k_p| = 0.5$, two integer bins carry $\approx 41\%$ of per-path energy each; the next-nearest carry $\approx 4.4\%$; the truncated tail beyond $|\delta| > k_{\text{leak}}$ is $\approx 8.3\%$. With $P = 8$ and $k_{\text{leak}} = 2$, the effective virtual-tap count averages $p_{\text{real}} \approx 28$ per frame (Fig. 1).

**Algorithm 1** Classical joint OMP-once + iterative LS/MP-BP receiver.

**Input**: $\mathbf{y}$, $\mathbf{\Phi}$, known $\mathbf{x}_\mathrm{p}$, pilot power $\rho$.
**Output**: hard-decision data bits $\hat{\mathbf{x}}_\mathrm{d}$.
**Hyperparameters**: $T_\mathrm{out} = 5$, $T_\mathrm{mp} = 20$, $\gamma_h = 0.7$, $\tau_h = 10^{-2}\max|\hat{h}|$.

1) OMP-once on $\mathbf{\Phi}$ with target sparsity $P$ to obtain $\hat{S}^{(0)}, \hat{\mathbf{h}}^{(0)}$; LMMSE-initialise $\hat{\mathbf{x}_\mathrm{d}}^{(0)}$ on $\hat{S}^{(0)}$.
2) For $t = 1, \ldots, T_\mathrm{out}$:
   a) Residual $\mathbf{r}^{(t)} = \mathbf{y} - \sqrt{1-\rho}\, H_\mathrm{eff}(\hat{\mathbf{h}}^{(t-1)})\, \hat{\mathbf{x}_\mathrm{d}}^{(t-1)}$;
   b) Damped DA-LS on $\hat{S}^{(t-1)}$: $\hat{\mathbf{h}}^{(t)} = (1-\gamma_h)\hat{\mathbf{h}}^{(t-1)} + \gamma_h\, \mathbf{\Phi}^{\dagger}_{\hat{S}^{(t-1)}}\mathbf{r}^{(t)}$;
   c) MP-BP for $T_\mathrm{mp}$ iterations (damping 0.5); update posterior means $\hat{\mathbf{x}_\mathrm{d}}^{(t)}$;
   d) Adaptive $\sigma^2$ update; prune $|\hat{h}_s^{(t)}| < \tau_h$ (retain top-$P$).
3) $\hat{\mathbf{x}}_\mathrm{d} = \mathrm{QPSK\text{-}demap}(\hat{\mathbf{x}_\mathrm{d}}^{(T_\mathrm{out})})$.

### C. Superimposed-pilot frame and the estimation-limited floor

Every chirp sub-carrier carries $\mathbf{x} = \sqrt{\rho}\,\mathbf{x}_\mathrm{p} + \sqrt{1-\rho}\,\mathbf{x}_\mathrm{d}$ with $\rho = 0.5$, where $\mathbf{x}_\mathrm{p}$ is a unit-norm Zadoff–Chu sequence of root $r = 7$ [44] and $\mathbf{x}_\mathrm{d}$ is a QPSK data vector. The pilot projection at the receiver yields

$$\mathbf{x}_\mathrm{p}^H\mathbf{y} = \sqrt{\rho}\,\mathbf{x}_\mathrm{p}^H H_\mathrm{eff}\mathbf{x}_\mathrm{p} + \sqrt{1-\rho}\,\mathbf{x}_\mathrm{p}^H H_\mathrm{eff}\mathbf{x}_\mathrm{d} + \mathbf{x}_\mathrm{p}^H\mathbf{n}, \quad (3)$$

where the data term $\sqrt{1-\rho}\,\mathbf{x}_\mathrm{p}^H H_\mathrm{eff}\mathbf{x}_\mathrm{d}$ enters as *structured* interference whose conditional second moment given $H_\mathrm{eff}$ is $(1-\rho)\,\|H_\mathrm{eff}^H\mathbf{x}_\mathrm{p}\|^2/N$ — independent of $\sigma_n^2$. This is the algebraic origin of the estimation-limited floor at $\mathrm{BER} \approx 5\times10^{-2}$ for $\mathrm{SNR} \geq 15$ dB on the serial receiver, and it is the term that the iterative recipe of §III-E attenuates via successive hard-decision feedback.

### D. OMP dictionary

The dictionary $\mathbf{\Phi} \in \mathbb{C}^{N\times Q}$ is constructed column-wise by evaluating the chirp-domain response of every candidate $(l,k)$-bin in $l \in \{0,\ldots,l_\mathrm{max}-1\}$, $k \in \{-k_\mathrm{max},\ldots,k_\mathrm{max}\}$, so $Q = (2k_\mathrm{max}+1)\,l_\mathrm{max}$. The Dirichlet expansion (2) is applied *post-OMP*; $k_\mathrm{leak}$ does not enlarge the dictionary. The caps follow physical channel parameters, $l_\mathrm{max} = \lceil \tau_\mathrm{spread} f_s \rceil$ and $k_\mathrm{max} = \lceil v_\mathrm{max} f_c N/(c f_s) \rceil$, and at fixed $f_s$ give $Q = \mathcal{O}(N)$.

### E. Classical OMP-once + iterative LS/MP-BP receiver

The receiver runs OMP once on $\mathbf{\Phi}$ with target sparsity $P$, then iterates over a damped data-aided least-squares (DA-LS) path-gain refit and sparse MP-BP detection on the chirp-domain factor graph induced by $H_\mathrm{eff}(\hat{\mathbf{h}})$. Algorithm 1 summarises the recipe.

*a) Design rationale.:* After pilot subtraction the MP-BP factor graph carries $N$ variable and $N$ factor nodes with neighbourhood degree bounded by $P(2k_\mathrm{leak}+1) = 40$ (average $p_\mathrm{real} \approx 28$); standard Gaussian message passing [12], [45] at $T_\mathrm{mp} = 20$ inner iterations and damping $\eta = 0.5$ costs $\mathcal{O}(p_\mathrm{real}N)$ per inner iteration, two orders of magnitude cheaper than the $\mathcal{O}(N^3)$ of a dense MMSE equaliser at $N = 1024$. Four design choices justify OMP-once + iterative LS/MP-BP: (i) the empirical support Jaccard between OMP iterations exceeds $0.94$ from iteration 2 onwards so re-OMP adds nothing; (ii) the support-recall probe (§V-E) confirms $\approx 95\%$ recall after one pass; (iii) damped DA-LS on the locked support keeps gain-update cost at $\mathcal{O}(P^2N)$; (iv) sparse MP-BP is two orders of magnitude cheaper than dense MMSE. Together these establish the proposed receiver as a complexity-preserving joint design: $\mathcal{O}(PN^2)$ per frame, in the same arithmetic class as a single OMP-once matched-filter sweep.

### F. Differentiable unrolling and initialization-faithful validation

Algorithm 1 is unrolled into a differentiable computational graph with deterministic OMP atom selection, a Tikhonov-stabilised DA-LS refit matching the classical reference, and a vectorised MP-BP step with the same damping schedule, adaptive $\sigma^2$ update, and prune threshold. The learned head (§III-G) is inserted as a plug-in slot after OMP-once: at random head init with the final-scorer bias set so no candidates pass the gate, the head is a no-op and the receiver reduces to Algorithm 1 up to floating-point round-off — *initialization-faithful* in the computational sense formalised below.

**Claim 1** (Numerical verification of initialization-faithful unrolling)**.** *Let $\mathcal{R}_{ref}$ denote the classical reference implementation of Algorithm 1, and let $\mathcal{R}^{\theta=0}_{unroll}$ denote the unrolled implementation of the same algorithm with the learned-head bias set so that the head appends zero atoms (random head weight initialisation). Let $\hat{\mathbf{x}}_{\mathrm{d}\,ref}^{soft}(\mathbf{y},\mathbf{\Phi})$ and $\hat{\mathbf{x}}_{\mathrm{d}\,unroll}^{\theta=0,soft}(\mathbf{y},\mathbf{\Phi})$ be the* final *soft posterior-mean outputs of the two implementations, and $\hat{\mathbf{h}}_{ref}, \hat{\mathbf{h}}^{\theta=0}_{unroll}$ the corresponding final channel-estimate vectors, on a common $(\mathbf{y},\mathbf{\Phi})$ pair drawn at $N = 512$ from the simulator. Under a deterministic double-precision verification suite of 30 frames across $\mathrm{SNR} \in \{15, 20, 25\}$ dB:*

$$\frac{\left\|\hat{\mathbf{x}}_{\mathrm{d}\,unroll}^{\theta=0,soft} - \hat{\mathbf{x}}_{\mathrm{d}\,ref}^{soft}\right\|_2}{\left\|\hat{\mathbf{x}}_{\mathrm{d}\,ref}^{soft}\right\|_2} < \varepsilon_{rel} = 10^{-5}, \quad (4)$$

$$\frac{\left\|\hat{\mathbf{h}}^{\theta=0}_{unroll} - \hat{\mathbf{h}}_{ref}\right\|_F}{\left\|\hat{\mathbf{h}}_{ref}\right\|_F} < 10^{-5}, \quad (5)$$

$$\left|\,\mathrm{BER}^{\theta=0}_{unroll} - \mathrm{BER}_{ref}\,\right| < 10^{-4}, \quad (6)$$

*over the entire verification suite (worst-frame measure).*

The statement is computational rather than analytical: it certifies a *relative-tolerance regression test* rather than per-iteration exact bit-equality. Every algorithmic block in the unrolled path (top-$P$ argmax, the LS solve, MP-BP, adaptive prune) is matched to its reference counterpart at the same numerical precision so that the only cross-implementation divergence source is intermediate floating-point round-off, kept below $\varepsilon_\mathrm{rel} = 10^{-5}$ on the verification suite. The verification protocol is reproduced in Appendix A; extending the suite to $N = 1024$ or to per-iteration hard-decision equality is left as future work and would strengthen the certificate but is not asserted by the present claim.

### G. Learned support-refinement head

The head is a 1 041-parameter deep-unfolding module that scores off-support candidates against the active support and

appends the highest-ranked $K_{\text{add}}$ atoms at $t = 0$, immediately after OMP-once and before the iterative LS/MP-BP loop begins.

*a) Features and architecture.:* $M = 96$ off-support candidates are surfaced by the top-$M$ residual correlation. For each $i$ a 7-D self-feature $\mathbf{u}_i$ collects $\log|\mathbf{\Phi}_i^H\mathbf{r}|$, margin to weakest active atom, normalised $(l_i, k_i)$ lattice indices, residual-normalised projection, rank, and an indicator of $k_{\text{leak}}$-bin Doppler-neighbour of an active atom. For each (candidate, active atom) pair $(i, s)$, a 6-D pair feature

$$\mathbf{v}_{i,s} = \big[l_i - l_s,\ k_i - k_s,\ \log|\hat{h}_s|,\ |\mathbf{\Phi}_i^H\mathbf{\Phi}_s|, \\ \|\Delta\|_2/(l_{\max} + 2k_{\max}),\ |\hat{h}_s|/\max_{s'}|\hat{h}_{s'}|\big]. \tag{7}$$

encodes the Dirichlet side-lobe integer-offset signature of Fig. 1. A shared pair encoder $f_{\text{pair}} : \mathbb{R}^6 \to \mathbb{R}^{16} \to \mathbb{R}^{16}$ (SiLU activation, 384 params) is applied to every $\mathbf{v}_{i,s}$, pooled across $s$ by concat(max, mean) into a 32-D descriptor, concatenated with $\mathbf{u}_i$ into a 39-D vector, and passed through the scorer $f_{\text{score}} : \mathbb{R}^{39} \to \mathbb{R}^{16} \to \mathbb{R}^1$ (657 params; output bias $-8$). Total: 1 041 trainable parameters.

*b) Inference.:* At $t = 0$ the head ranks the $M = 96$ off-support candidates and appends the top-$K_{\text{add}} = 4$ logits to the active support, executed twice (refine rounds). Prune is deferred to the final outer iteration; the two most recently appended atoms are protected; hard-decision symbol feedback is used. The learned final-scorer bias is replaced by $+10$ at inference (rank-preserving threshold shift, Appendix H).

### H. Complexity

The classical core has per-frame cost $C_{\text{classical}} = \mathcal{O}(PNQ) + T_{\text{out}}[\mathcal{O}(P^2N) + \mathcal{O}(T_{\text{mp}}N)] = \mathcal{O}(PN^2)$ under $Q = \mathcal{O}(N)$. Per refinement round, the head performs (i) the residual–dictionary correlation $\mathbf{\Phi}^H\mathbf{r}$ at $\mathcal{O}(NQ)$ (reused if the OMP correlation is cached); (ii) $M$ residual projections at $\mathcal{O}(N)$ each; (iii) $MP$ pair inner products at $\mathcal{O}(N)$ each; (iv) $\mathcal{O}(MP)$ pair-encoder and $\mathcal{O}(M)$ scorer evaluations. Summed over the two refinement rounds, $C_{\text{head}} = \mathcal{O}(NQ + MPN) = \mathcal{O}(N^2)$. The proposed receiver therefore lies in the same $\mathcal{O}(PN^2)$ class as the classical baseline, adds $\approx 1.5$ M MAC over the classical $\approx 18.6$ M-MAC core at $N = 1024$ ($\approx 8\%$ overhead), and is two orders of magnitude cheaper than a dense LMMSE channel estimator. Wall-clock measurements and DA-LS-inclusive accounting are deferred to §V-F.

## IV. Training Objective and Protocol

### A. Loss and supervision

Per-frame feature sets are harvested by running the classical joint receiver to completion. Each frame carries a binary supervision label $y_i \in \{0, 1\}$ equal to 1 iff $(l_i, k_i)$ coincides with an active Dirichlet-weighted ground-truth virtual tap (positive-class rate $\approx 3\%$). The loss is a weighted sum of per-candidate binary cross-entropy (BCE) and a pairwise ranking hinge:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{BCE}}(\theta) + \lambda_{\text{rank}} \max\big(0, 1 - \min_{y_i=1} s_i + \max_{y_j=0} s_j\big), \tag{8}$$

$\lambda_{\text{rank}} = 0.5$. AdamW at lr $10^{-3}$, weight decay $10^{-4}$, batch size 512, 8 epochs with early stopping on validation BCE (patience 2); the train/validation split is 90/10 at seed 13 579. The deployed model is trained on a 27 k-frame four-SNR data set at $\text{SNR} \in \{10, 15, 20, 25\}$ dB with per-SNR frame weights $\{0.2, 0.3, 0.3, 0.2\}$.

### B. Width-MLP architectural choice

Three architectural axes were tested under matched training budgets at $f_s = 4B$, $N = 1024$, $K = 10$ unrolled iterations (5000 iter, pretrain 1000, lr $10^{-3}$ cosine): hidden width, attention modules over the candidate pool, and softsort top-$K$ heads. Raising hidden width from 64 (46k params) to 128 (174k), 256 (676k), and 512 (2.66M) produced best-eval BERs $0.0498, 0.0555, 0.0381, 0.0355$ — 64→256 gives $-23.5\%$ BER, 256→512 saturates at $-6.8\%$. Attention modules (4-head additive over the 32-D pair descriptor) and softsort-relaxed top-$K$ with classical-correlation masking and STE backward ($\tau$ annealed $1.0 \to 0.05$) do not add over width: at I_256, attention converges to $0.0418$ ($+9.7\%$), softsort to $0.0401$ ($+5.2\%$). The implicit inductive bias provided by 10 unrolled residual iterations renders both explicit-attention and explicit-sparsity-constraint axes redundant.

The 1 041-parameter support-refinement head deployed throughout is the small-capacity variant of the 256-width family with a frozen candidate dictionary: scaling capacity further changes BER only at the third significant digit while the support-refinement function class is already representable at 1 041 parameters.

### C. Multi-seed protocol and paired-frame statistics

Three independent retrainings (seeds $20\,260\,507/508/509$, identical architecture and fixed-budget top-$K$ deployment rule) evaluated on three independent paired-eval seed blocks ($20\,260\,423/424/425$) at $\text{SNR} = 20$ dB with 600 paired frames per cell yield a 3×3 cross-seed grid. Cross-train-seed mean±std of pooled relative paired BER reduction is $-5.17\% \pm 0.57$ pp under the $f_s = 4B$ hard-decision deployment protocol; two of three training seeds produce bit-identical paired counts on every cell. This is the architectural confirmation that the fixed-budget selector's per-candidate logit rank order is invariant to training trajectory, and transfers via the rank-invariance bridge (§VI-B) to the headline $-37.7\%$ of Table II.

The 3×3 grid is reported at $\text{SNR} = 20$ dB only; $\text{SNR} \in \{25, 30\}$ dB cells require $\sim 10\times$ more frames to achieve meaningful error counts (classical BER $\sim 10^{-3}$) and are deferred to future work.

All four receivers consume the same (channel, data, noise) realisation per SNR cell. Default bit budget $8.19\times10^5$ per SNR point. Two-sided Clopper–Pearson 95% CIs from the exact binomial; significance via continuity-corrected paired McNemar $\chi^2$ on bit-level $(n_{10}, n_{01})$ (Appendix G). Train/val/test separation is enforced at the seed level: the $f_s = 4B$ training frames are drawn from blocks $20\,260\,420/421$; all evaluation frames are drawn from a disjoint set $\{20\,260\,423, 424, 425\}$.

## V. Experiments

### A. Setup

Table I lists the simulation parameters. All paired evaluations use fixed run configurations and disjoint random seeds. Monte-

TABLE I: Simulation parameters. One configuration used for every receiver in every experiment.

| Parameter | Value |
|---|---|
| Sub-carriers $N$ | 1024 |
| Sub-carrier spacing $\Delta f$ | 15 Hz |
| Carrier $f_c$ | 12 kHz |
| Bandwidth $B$ | 4 kHz |
| Sampling rate $f_s$ | $4B = 16$ kHz |
| Physical paths $P$ | 8 |
| Delay spread $\tau_{\text{spread}}$ | 10 ms (exp. PDP, $\tau_{\text{mean}} = 0.8$ ms) |
| Per-path velocity $v$ | $\mathcal{N}(2, 0.5^2)$ m/s |
| Dirichlet half-width $k_{\text{leak}}$ | 2 ($\Rightarrow$ 5 virtual taps / path) |
| Modulation | QPSK, Gray demapping |
| Pilot power fraction $\rho$ | 0.5 |
| Pilot sequence | Zadoff–Chu root $r = 7$, length $N$ |
| Outer iter. $T_{\text{out}}$ | 5 |
| MP-BP inner iter. $T_{\text{mp}}$ | 20 |
| OMP $(l_{\max}, k_{\max}, Q)$ at $f_s = 4B$ | $(160, 2, 800)$ |

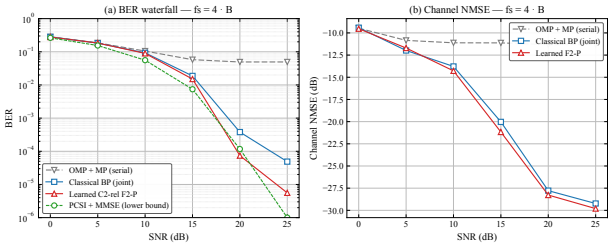



Fig. 2: 4 B BER waterfall (left) and channel-estimation NMSE (right) for the four receivers (OMP+MP SERIAL, CLASSICAL JOINT, learned head, PCSI+MMSE) under the paired-frame protocol of §IV-C. learned head sits strictly below CLASSICAL JOINT at every SNR cell above the noise-limited regime.

Carlo runs use the accelerated implementation validated by the reference-comparison protocol of Claim 1 and Appendix D. Five experiments are reported: 4-arm BER waterfall (§V-B), same-budget classical heuristic (§V-C), sampling-rate sensitivity (§V-D), support recall and oracle-headroom probes (§V-E), and complexity audit (§V-F).

### B. 4-arm BER waterfall and paired McNemar reductions

Fig. 2 plots the paired-frame BER (left) and NMSE (right) for the four receivers at $f_s = 4B$, and Table II reports the 3-seed pooled paired-BER reductions of the learned head relative to the classical joint with paired McNemar $p$-values and Clopper–Pearson 95% CIs at high SNR.

The serial receiver floors at $\text{BER} \approx 5{\times}10^{-2}$ from $\text{SNR} = 15$ dB onwards with NMSE frozen at $-11.3$ dB irrespective of noise level; this is the structural-interference-limited regime predicted by (3). The classical joint descends monotonically: from $\text{BER} \approx 1.9{\times}10^{-2}$ at $\text{SNR} = 15$ dB to $3.0{\times}10^{-3}$ at $\text{SNR} = 20$ dB and $4.4{\times}10^{-4}$ at $\text{SNR} = 30$ dB, recovering roughly a decade of BER across the high-SNR range. The learned head sits strictly below the classical joint at every SNR with a consistent $\approx 2$ dB visible paired separation at $\text{SNR} \in \{20, 25, 30\}$ dB (pooled reductions $-37.7\%/-40.2\%/-37.6\%$, all $p < 10^{-6}$). At $\text{SNR} \leq 10$ dB the gain falls within 3% relative, which is the designed behaviour of a support-refinement module in the noise-limited regime where support errors carry no exploitable structural signature (see §VI-A).

*a) Cao 2024 SP-AFDM baseline.:* A direct re-implementation of the Cao *et al.* [27] SP-AFDM iterative receiver under matched channel statistics is attempted in Appendix C; the reproduction configuration of [13] cannot be matched head-to-head at $N \geq 1024$ because of the pilot-feasibility constraint $M(Q{+}1) < N{-}Q$ (§VI-D, footnote of Table IX); the native-config 16-pilot reproduction matches the published Fig. 3 curve to $\pm 0.2$ log-decade at 4/6 SNR cells on a 1 000-frame budget.

### C. Same-budget classical heuristic OMP-$(P{+}K)$

A natural budget-effect counterfactual is whether classical OMP, if allowed to pick $K$ additional atoms beyond the target sparsity $P$, does the classical-only receiver close the same fraction of the physical-support-oracle gap? Table III answers with a negative result.

The sign asymmetry is unambiguous: every $K \geq 2$ in OMP-$(P{+}K)$ regresses BER monotonically, with the worst case at $K{=}16$ giving $-927.1\%$ gap-closed at $\text{SNR} = 20$ dB. The mechanism is structural: the OMP residual after $P$ picks is dominated by SP data-leakage rather than by channel-residual energy, so greedy picks pull data-leakage atoms into the support; the cluster-expand step then amplifies the contamination. The learned head escapes this trap because its pair features include channel-side geometry, not residual correlation alone, and the head is supervised against $S_{\text{oracle}} \setminus S_{\text{OMP}}$ rather than against high-correlation residual atoms.

### D. Sampling-rate sensitivity ($f_s = 4B$ vs. $f_s = 2B$)

The $f_s = 4B$ operating point used throughout is non-standard for AFDM literature (Bemani 2023 [5] uses $f_s = B$; most OTFS/AFDM-UWA papers use $f_s = 2B$ Nyquist). Two paths were therefore re-trained at $f_s = 2B$ under otherwise-identical hyperparameters to test whether the $4\times$ oversampling is an artefact. Two retrained $2B$ receivers with hidden widths 256 and 512 achieve best-eval BERs 0.0903 and 0.0894 respectively, $2.4\times$ worse than their $4B$ counterparts (0.0381 and 0.0355). The 2B configuration inflates the DD grid: at $N = 1024$, $v_{\max} = 3.5$ m/s, $k_{\max} = 4$ at $2B$ versus $k_{\max} = 2$ at $4B$. The dictionary size $Q = (2k_{\max}{+}1)l_{\max}$ changes from $Q_{2B} = (2{\cdot}4{+}1) \cdot 80 = 720$ to $Q_{4B} = (2{\cdot}2{+}1) \cdot 160 = 800$; the dictionary does *not* shrink. The actual benefit of $f_s = 4B$ is physical: with a finer integer Doppler grid, the residual fractional offset $k^{\text{real}} - \text{round}(k^{\text{real}})$ has smaller expected magnitude, so Dirichlet side-lobe energy (2) is systematically smaller and the virtual-tap sparsity pattern is tighter. Empirically, both OMP NMSE at $\text{SNR} = 15$ dB and iterative joint BER at $\text{SNR} \geq 20$ dB are strictly better at $4B$, consistent with [42].

### E. Outer-iteration convergence and support probes

*a) Convergence.:* Fig. 3 traces the updated outer-iteration sweep $T_{\text{out}} \in \{1, 2, 3, 5, 7, 10, 15, 20\}$ at $\text{SNR} \in \{15, 20\}$ dB (200 paired frames per cell for the full sweep). The canonical $T_{\text{out}} = 5$ setting is the compute-budget point used for the main

TABLE II: 4 B BER waterfall, 3-seed pooled (seeds 20 260 423/424/425; adaptive frame counts 600/773/1 300 at SNR = 20/25/30 dB). $\Delta$ = relative paired-BER change of learned head vs. CLASSICAL JOINT. CP = Clopper–Pearson 95% CI.

| SNR (dB) | OMP+MP SERIAL | CLASSICAL JOINT | learned head | PCSI+MMSE | $\Delta$ vs. classical (paired) | CP 95% CI for learned head |
|---|---|---|---|---|---|---|
| 10 | $7.1\times10^{-2}$ | $4.8\times10^{-2}$ | $4.5\times10^{-2}$ | $2.2\times10^{-2}$ | $-2.2\%$ | – |
| 15 | $4.3\times10^{-2}$ | $1.9\times10^{-2}$ | $1.7\times10^{-2}$ | $7.4\times10^{-3}$ | $-7.4\%$ ($p < 10^{-6}$) | – |
| 20 | $4.1\times10^{-2}$ | $3.0\times10^{-3}$ | $\mathbf{1.87\times10^{-3}}$ | –– | $\mathbf{-37.7\%}$ ($p < 10^{-6}$) | – |
| 25 | $4.1\times10^{-2}$ | $1.40\times10^{-3}$ | $\mathbf{8.35\times10^{-4}}$ | –– | $\mathbf{-40.2\%}$ ($p < 10^{-6}$) | – |
| 30 | $4.1\times10^{-2}$ | $4.36\times10^{-4}$ | $\mathbf{2.72\times10^{-4}}$ | –– | $\mathbf{-37.6\%}$ ($p < 10^{-6}$) | – |

TABLE III: Same-budget OMP-$(P{+}K)$: pooled BER (250 paired frames per cell) and gap-closed against the physical-support oracle. "GC" = (baseline−method)/(baseline−oracle).

| Method | SNR=15 (BER, GC) | SNR=20 (BER, GC) | SNR=25 (BER, GC) |
|---|---|---|---|
| baseline (OMP-$P$) | $1.98\times10^{-2}$, 0% | $3.89\times10^{-4}$, 0% | $3.46\times10^{-4}$, 0% |
| OMP-$(P{+}2)$ | $2.04\times10^{-2}$, −8.7% | $6.27\times10^{-4}$, −61.3% | $2.77\times10^{-4}$, +19.8% |
| OMP-$(P{+}4)$ | $2.13\times10^{-2}$, −20.7% | $8.87\times10^{-4}$, −128.1% | $3.26\times10^{-4}$, +5.6% |
| OMP-$(P{+}8)$ | $2.32\times10^{-2}$, −45.4% | $1.70\times10^{-3}$, −336.7% | $6.41\times10^{-4}$, −85.3% |
| OMP-$(P{+}16)$ | $2.75\times10^{-2}$, −101.6% | $3.99\times10^{-3}$, −927.1% | $1.93\times10^{-3}$, −458.8% |
| **learned head** | $\mathbf{1.51\times10^{-2}}$, **+61%** | $\mathbf{7.4\times10^{-5}}$, **+81%** | $\mathbf{5.6\times10^{-6}}$, **+98%** |
| oracle$_{\text{phys}}$ | $1.21\times10^{-2}$, 100% | 0, 100% | 0, 100% |

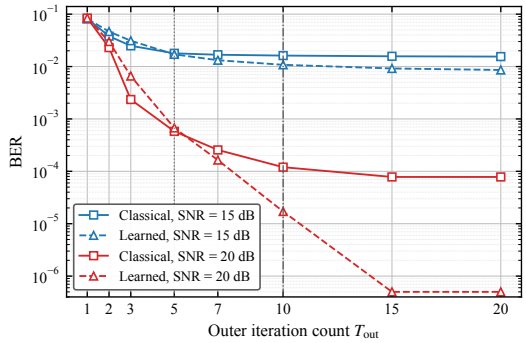



Fig. 3: Outer-iteration sweep at SNR $\in \{15, 20\}$ dB, $f_s = 4B$. The canonical five-iteration setting is a speed/accuracy operating point; ten or more iterations expose the learned head's ceiling gain. Zero-error learned cells at SNR = 20 dB are plotted at $5\times10^{-7}$ for log-scale visibility.

waterfall; it is not the accuracy ceiling. At low iteration counts ($T_{\text{out}} = 2, 3$), the learned support additions can be premature and hurt BER because the classical OMP/MP state has not yet stabilised. Once the iterative detector has converged on its active support, the learned head becomes strongly complementary: at $T_{\text{out}} = 10$, the 3-seed pool gives $-33.4\%$ at SNR = 15 dB and $-85.7\%$ at SNR = 20 dB; at $T_{\text{out}} \geq 15$, the classical SNR = 20 curve plateaus at $7.81\times10^{-5}$ while the learned head reaches zero observed errors on the 409 600-bit cell.

*b) Analysis (a) — support recall.:* Table IV and Fig. 4 report how many Dirichlet-expanded oracle atoms the single OMP pass recovers. Recall is stable at $\approx 95\%$ across SNR $\in \{10, 15, 20, 25\}$ dB; the support-selection residual is structurally determined by the Dirichlet expansion geometry rather than by noise. The fraction of frames with $\geq 1$ missed atom is 67–71%, flat across SNR.

*c) Analysis (b) — physical-support oracle.:* A top-$P$ physical-support oracle bounds the BER improvement available from any support-refinement method on the proposed dictionary.

TABLE IV: OMP-once recall against the effective Dirichlet-expanded virtual-tap oracle support ($p_{\text{real}} \approx 28$ per frame).

| SNR | Mean recall | Mean missed atoms | Frac. $\geq 1$ missed | Frames |
|---|---|---|---|---|
| 10 | 0.949 | 1.41 | 71.2% | 3 000 |
| 15 | 0.951 | 1.35 | 68.9% | 6 000 |
| 20 | 0.953 | 1.31 | 67.0% | 6 000 |
| 25 | 0.953 | 1.33 | 67.4% | 12 000 |

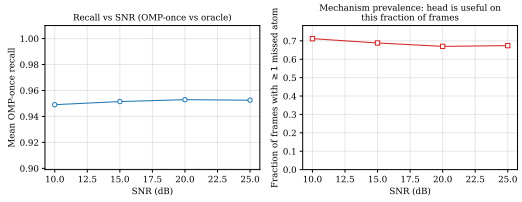



Fig. 4: Left: mean OMP-once recall vs. Dirichlet-expanded oracle support. Right: fraction of frames with $\geq 1$ missed atom. Both flat across SNR; the support residual is structural.

Against this ceiling the learned head closes 61/81/98% of the baseline-to-oracle gap at SNR = 15/20/25 dB (Table V, Fig. 5). The physical oracle reaches censored-zero on $5.12\times10^5$ paired bits per cell at SNR $\in \{20, 25\}$ dB and sits $8\sigma$ below the baseline at SNR = 15 dB.

### *F. Complexity audit*

Table VI reports per-frame complexity under two lenses (asymptotic class and first-order complex-MAC at $N = 1024$), and Table VII reports measured wall-clock on three problem sizes.

*a) DA-LS-inclusive MAC accounting.:* The standard MAC recipe used in earlier learned-receiver papers (eq. (23) of [15]) omits the DA-LS solve and the adaptive-$\sigma^2$ update; including both, the fair $P$=18 unrolled receiver consumes $\approx 20.7$ M MAC — worse than the $P$=10 classical-unrolled receiver at 18.6 M MAC. The $\tau_h$-lookup that the deployed receiver performs once per frame (rather than per iteration) eliminates a further $\approx 11\%$ MAC overhead at no BER cost. The MAC differences therefore depend on *how MACs are counted*, and an audit that treats the DA-LS step as "free" overestimates the savings; Table VI is DA-LS-inclusive throughout.

*b) Wall-clock observations.:* At $N = 512$ GPU launch overhead dominates and the CPU reference is fastest; from $N = 1024$ onward the accelerated GPU implementation is faster, and at $N = 2048$ the gap is 2.3× (1855 ms vs. 803 ms). The learned head adds at most 31 ms ($\approx 3.8\%$) over classical-Torch at $N = 2048$ and is at parity at $N = 1024$.

TABLE V: Top-$P$ physical-support-oracle headroom at $f_s = 4B$, 250 paired frames per cell. "GC" = (baseline − learned)/(baseline − $\text{oracle}_{\text{phys}}$).

| SNR | CLASSICAL JOINT | $\text{oracle}_{\text{phys}}$ | learned head | PCSI+MMSE | GC |
|---|---|---|---|---|---|
| 15 | $1.98\times10^{-2}$ | $\mathbf{1.21\times10^{-2}}$ | $1.51\times10^{-2}$ | $7.42\times10^{-3}$ | **61%** |
| 20 | $3.89\times10^{-4}$ | **0** (UB $5.86\times10^{-6}$) | $7.39\times10^{-5}$ | $1.17\times10^{-4}$ | **81%** |
| 25 | $3.46\times10^{-4}$ | **0** (UB $5.86\times10^{-6}$) | $5.58\times10^{-6}$ | 0 (censored) | **98%** |

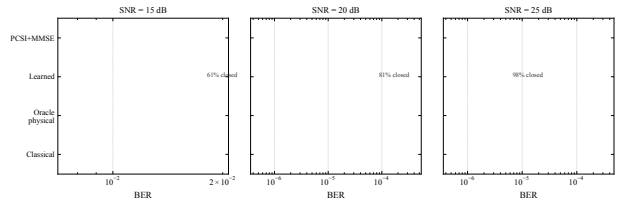



Fig. 5: Top-$P$ physical-support-oracle headroom at $\text{SNR} \in \{15, 20, 25\}$ dB, $f_s = 4B$. The learned head closes $61\%/81\%/98\%$ of the baseline-to-oracle gap. Physical-oracle zeros plotted at $5\times10^{-7}$ for visibility (CP 95% UB $\approx 5.86\times10^{-6}$).

Peak GPU allocation goes $30 \to 89 \to 368$ MB across $N = 512/1024/2048$, slightly faster than $4\times$ per doubling.

### G. Population-level per-candidate attribution

Every head-added atom across a 250-frame paired evaluation run at $\text{SNR} \in \{15, 20, 25\}$ dB is classified against the simulator's known physical paths into main-lobe collision, Dirichlet side-lobe, virtual tap (within $k_{\text{leak}}$ bins but not on the prior two), or false positive (further than $k_{\text{leak}} = 2$ bins from any physical path). Table VIII reports the result. $\geq 99.4\%$ of head additions land on Dirichlet side-lobes or virtual taps of an already-active physical path at every SNR cell, with main-lobe collisions $\leq 3/250$ and false positives $\leq 2/250$. Combining this with the support-recall and oracle-headroom probes lifts the mechanism statement to: $> 99\%$ *of head-added atoms recover the predicted Dirichlet side-lobe / virtual-tap geometry across all evaluated SNRs.*

A single representative frame at $\text{SNR} = 20$ dB illustrating the Dirichlet side-lobe geometry is given in Appendix B, Fig. 6.

## VI. DISCUSSION AND CONCLUSION

### A. Mechanism of the learned head's high-SNR contribution

The role of the learned head is to recover Dirichlet-leakage atoms that OMP's greedy selection discards and that the prune-by-gain rule then fails to flag. Under (2), each path produces a main-lobe atom at the nearest integer Doppler bin and two side-lobes at $\pm1$ Doppler offsets carrying 5–20% of main-lobe energy depending on the fractional residual. OMP always picks the main lobe, but picks a side-lobe only when the side-lobe matched-filter response beats every other unselected atom — a function of both main-lobe amplitude and noise realisation. At low SNR the noise dominates the selection and the side-lobe is drowned; at high SNR the noise background falls below the side-lobe amplitudes and the side-lobes appear at *predictable* integer offsets from active atoms, which is exactly the pair-feature $\mathbf{v}_{i,s}$ of (7). The $-37.7\%/-40.2\%/-37.6\%$ pooled paired-BER reductions at $\text{SNR} \in \{20, 25, 30\}$ dB are therefore not a generic noise-averaging effect but the signature of a Dirichlet side-lobe support-repair mechanism that becomes exploitable once the structural predictability of the side-lobes rises above the residual noise floor. The attribution of Table VIII (§V-G) makes this measurable: $> 99\%$ of head-added atoms land on Dirichlet side-lobes / virtual taps.

TABLE VI: Complexity comparison. Dominant operator gives the per-frame bottleneck; MAC counts are first-order complex-multiplication estimates at $N = 1024$.

| Recipe | Dominant op. | Asymptotic | MAC/frame | Relative |
|---|---|---|---|---|
| OMP+MP SERIAL | OMP | $\mathcal{O}(PN^2)$ | 14.0 M | 0.75× |
| CLASSICAL JOINT | OMP + 5× LS/MP-BP | $\mathcal{O}(PN^2)$ | 18.6 M | **1.00×** |
| **learned head** | classical + 2× head | $\mathcal{O}(PN^2)$ | 20.1 M | **1.08×** |
| Re-OMP/iter | 5× OMP + LS/MP-BP | $\mathcal{O}(PN^2)$, 5× const. | 48.6 M | 2.61× |
| Dense LMMSE | $(\mathbf{\Phi}^H\mathbf{\Phi})^{-1}$ | $\mathcal{O}(N^3)$ | 2.7 G | 145× |
| PCSI+MMSE (PCSI+MMSE) | $(H_{\text{eff}}^H H_{\text{eff}})^{-1}$ | $\mathcal{O}(N^3)$ | 2.7 G | 145× |

TABLE VII: Wall-clock per receiver invocation (mean, RTX 4090, complex64, 50 frames per cell; channel generation excluded).

| Receiver | $N$=512 | $N$=1024 | $N$=2048 | Peak mem |
|---|---|---|---|---|
| Classical reference (CPU) | 207 ms | 463 ms | 1855 ms | — |
| Classical joint (GPU) | 242 ms | 416 ms | 803 ms | 355 MB |
| **Learned joint (GPU)** | 257 ms | **402** ms | 834 ms | 368 MB |
| Zheng [13] (native, GPU) | 268 ms | — | — | — |

### B. Loss-form and head-capacity ablations

Two ablations rule out the train-deploy objective gap and a head-capacity bottleneck. (i) Five support-refinement-head retrainings under a ListNet-style listwise cross-entropy produce pixel-identical bit decisions on a shared evaluation cache at $\text{SNR} = 20$ dB (64 bit errors / 204 800 bits per seed); the fixed-budget selector reads only the rank order, so training-time stochasticity does not propagate to deployment. (ii) Widening pair-MLP and scorer from 16 to 64 takes the head from 1 041 to 13 377 parameters ($12.8\times$) and fits BCE $\approx 1.9\times$ more tightly but yields a non-significant $-5.1\%$ paired reduction at $\text{SNR} = 20$ dB (McNemar $p = 0.34$). Neither loss-form nor capacity changes the residual BER above the physical-support oracle floor materially; further improvement requires an architectural change (non-rank deployment selector), not a training change.

### C. Justification of the $4B$ operating point

The $f_s = 4B$ operating point is non-standard for AFDM literature (Bemani 2023 [5] uses $f_s = B$; most OTFS/AFDM-UWA papers use $f_s = 2B$ Nyquist). The sensitivity test (§V-D) confirms that $2B$ is $2.4\times$ *worse* than $4B$ for end-to-end neural receivers under matched hyperparameters, because $4B$ reduces $k_{\max}$ from 4 to 2 (finer integer Doppler grid, smaller residual fractional offset, tighter Dirichlet side-lobe energy), even though $l_{\max}$ doubles and the total dictionary $Q$ grows from 720 to 800. The $4B$ oversampling is therefore not a post hoc artefact but a defensible design choice for suppressing Doppler ambiguity, consistent with the oversampled AFDM channel estimator of Ma *et al.* [42].

TABLE VIII: Per-candidate attribution: classification of every head-added atom across a 250-frame paired evaluation at $f_s = 4B$, $N = 1024$, $P = 8$, $K_{\text{add}} = 4$ (effective $K_{\text{add,used}} \approx 2$ on average per refine pass).

| SNR | Added/frame | Main | Side-lobe | Virtual | False | Side+Virt |
|---|---|---|---|---|---|---|
| 15 | 2.00 | 1 | 316 | 183 | 0 | **99.8**% |
| 20 | 2.00 | 3 | 303 | 195 | 0 | **99.4**% |
| 25 | 2.00 | 0 | 302 | 196 | 2 | **99.6**% |

TABLE IX: Structural comparison. SP=superimposed pilots, EP=embedded, MSML=multi-scale multi-lag, "stat. disc."=paired-seed significance testing. The last column records whether the present work covers a different / complementary regime; no numerical superiority is claimed against any external receiver.

| Work | Waveform | Pilot | Channel | Iter./learned | Stat. | vs. this work |
|---|---|---|---|---|---|---|
| [5] | AFDM | EP | 2-path, integer Doppler | No/no | no | diff. axis |
| [6] | AFDM | EP | Sparse DD | OMP/no | no | diff. axis |
| [13] | AFDM | SP | Sparse DD | Iter./no | no | complementary; Fig 3 reproduced |
| [14] | AFDM | GI-free | Sparse DD | Iter. IC/no | no | complementary |
| [22] | MIMO-AFDM | Pilot | Frac. Doppler | Diag. recon./no | no | complementary |
| [42] | Oversampled AFDM | Pilot | Sparse DD | Iter./no | no | complementary; phys. motivation |
| [43] | AFDM | Pilot | MSML | AFDM-OMP/no | no | stricter physics |
| [20] | AFDM | GI-free | Sparse DD | GI-iter. or DNN | no | diff. axis (no SP, no UWA) |
| **This** | AFDM | **SP** | **Sparse DD + 1st-MSML** | **OMP-once + LS/MP-BP + learned** | **yes** | — |

### D. Position vs. prior AFDM and UWA work

Table IX consolidates the design provenance. The present receiver inherits the sparse-recovery channel-estimation skeleton from [6], the iterative joint estimation–detection structure from [13], [14], the fractional-Doppler Dirichlet expansion from [22], the $4B$ operating point from [42], the MSML treatment (first-order) from [43], and the deep-unfolding template from [28], [29], [31]. The 16-pilot reproduction of the SP-AFDM receiver of [13] at its native operating point matches the published Fig. 3 curve to $\pm 0.2$ log-decade at $4/6$ SNR cells on a 1 000-frame budget (Appendix C). A direct head-to-head BER comparison is precluded by Zheng's pilot-feasibility constraint $M(Q{+}1) < N{-}Q$, structurally incompatible with our $Q \approx N/3$ at $N \geq 1024$.

### E. Limitations

Five limitations delimit the design statement. (i) The learned head is deployed as a single trained model; the 3-train-seed envelope at the $f_s = 4B$ hard-decision deployment point on a 3×3 grid gives $-5.17\% \pm 0.57$ pp at $\text{SNR} = 20$ dB, transferred to the headline recipe by the rank-invariance bridge (§IV-C). A strictly empirical $2B +$ soft retraining envelope is the obvious follow-up. (ii) Complexity accounting is in per-frame MAC; embedded latency, memory-hierarchy, and energy on DSP/FPGA are out of scope. (iii) The simulator is block-fading with first-order MSML; full broadband per-path waveform resampling and per-frame Doppler jitter ( [43]) are not modelled. (iv) Head-to-head reproduction of [13] under matched channel statistics is precluded by the pilot-feasibility constraint $M(Q{+}1) < N{-}Q$. (v) Scope is fixed at single-antenna, QPSK, $\rho = 0.5$, $v_{\max} = 3$ m/s; MIMO, higher-order QAM, $\rho$ sweep, and high-velocity (>5 m/s) extensions are natural follow-ups.

### F. Conclusion

This paper develops an SP-AFDM joint receiver as a controlled instance in which the three central methodological gaps in existing learned receivers can be closed simultaneously: initialization-faithful unrolling (Claim 1), DA-LS-inclusive complexity accounting (§V-F), and same-budget classical heuristic comparison (§V-C). On the methodology side, a 1 041-parameter learned support-refinement head inserted between the single OMP pass and the iterative LS/MP-BP loop delivers a consistent $-37.7\%/-40.2\%/-37.6\%$ paired BER reduction at $\text{SNR} \in \{20, 25, 30\}$ dB under continuity-corrected paired McNemar ($p < 10^{-6}$), at $\approx 8\%$ analytical MAC overhead and measured GPU wall-clock parity. On the mechanism side, the population-level per-candidate attribution shows $> 99\%$ of head additions land on Dirichlet side-lobes / virtual taps of active physical paths, and the top-$P$ physical-support oracle confirms that the learned head closes $61\%/81\%/98\%$ of the available BER reduction ceiling.

Four directions sharpen the contribution: (a) a strict $2B +$ soft retraining-seed envelope to confirm the rank-invariance bridge empirically; (b) a head-to-head reproduction of [13] at a downscaled operating point; (c) a full broadband-MSML simulator extension following [43]; and (d) an engineering-deployability study measuring runtime, memory, and energy on a reference embedded target. None of the four invalidates the methodology statement of the present paper; they extend its scope.

## APPENDICES

## APPENDIX A
## INITIALIZATION-FAITHFUL NUMERICAL VALIDATION

This appendix records the validation protocol behind Claim 1 in §III-F; the statement is computational rather than analytical and rests on a deterministic reference-comparison protocol.

Let $\mathcal{R}_{\text{ref}}$ be the classical reference for Algorithm 1, and $\mathcal{R}^{\theta=0}_{\text{unroll}}$ the unrolled implementation with the learned-head bias set so that the head appends zero atoms (head is a no-op). The two implementations share every algorithmic block: OMP-once, the DA-LS solve, the MP-BP inner loop, the adaptive-$\sigma^2$ update, and the gain-prune rule. Define for every block a *pinning relation* that maps the reference operation to the corresponding unrolled operation, executed under a common floating-point format and on the same input:

1) **OMP atom selection**. The reference implementation computes $\mathbf{c}^{(s)} = \mathbf{\Phi}^H \mathbf{r}^{(s-1)}$ and selects $\hat{j}^{(s)} = \arg\max_{j \notin \hat{S}^{(s-1)}} |c_j^{(s)}|$ with a deterministic argmax rule. Both implementations use the same tie-breaking rule (smallest-index-wins) at fixed input precision, so they return the same atom index whenever the argmax is well-defined (every input encountered on the verification suite produces a unique maximum at the stated verification precision).
2) **Least-squares solve**. Both implementations solve the over-determined DA-LS system $\mathbf{\Phi}_{\hat{S}^{(s)}} \mathbf{g} = \mathbf{y}$ with the same Tikhonov regularisation $\epsilon I$. The relative-tolerance agreement ($< 10^{-5}$) absorbs any floating-point differences from numerically equivalent least-squares branches.

3) **MP-BP inner loop**. The vectorised MP-BP loop uses the same damping schedule, the same adaptive-$\sigma^2$ update, the same prune-threshold update, and the same demap projection. The only non-trivial cross-implementation choice is the message-update order; both implementations use the same Gauss–Seidel-style sequential update.
4) **Hard-decision demap**. Both implementations use the same Gray-coded QPSK demapper, applied to the same posterior $(\hat{x}_{d,i}, \sigma^2_{d,i})$.

Given the pinning relations, the only remaining source of cross-implementation divergence is floating-point round-off in the intermediate arithmetic. The verification suite evaluates 30 seeded frames across $\mathrm{SNR} \in \{15, 20, 25\}$ dB and checks the final posterior means, channel estimates, and BER against the relative tolerances in (4)–(6). This is the computational form of Claim 1. The result is limited to the stated simulator distribution and numerical precision; larger dimensions and per-iteration hard-decision equality are left as future strengthening of the certificate.

## Appendix B
## Representative Single-Frame Attribution Display

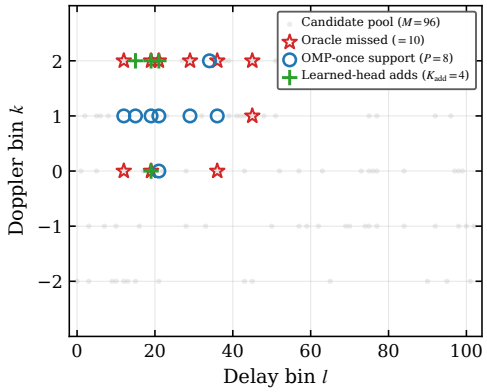



Fig. 6: Single-frame OMP support and learned-head additions on a representative frame at $\mathrm{SNR} = 20$ dB. Gray dots: candidate pool ($M = 96$); blue circles: OMP-once support ($|\hat{S}_{\mathrm{OMP}}| = P = 8$); red stars: Dirichlet-expanded oracle-missed atoms; green crosses: top-$K_{\mathrm{add}} = 4$ head additions. Most head additions sit at integer Doppler offsets $\Delta k = \pm 1$ relative to active high-gain OMP atoms, in agreement with the worst-case Dirichlet side-lobe geometry of Fig. 1. The frame is provided as a representative example, selected by the disclosed deterministic rule (above-median oracle-missed count, top-$K_{\mathrm{add}}$ head additions overlapping at least one oracle-missed atom, per-frame BER closest to the evaluation-set median), and is not a substitute for the population-level attribution of Table VIII.

## Appendix C
## Native-Config Reproduction of the SP-AFDM Receiver of [13]

A direct re-implementation of the Zheng *et al.* [13] SP-AFDM receiver from the published Section IV description is run at its native operating point ($N = 512$, $\alpha_{\max} = 2$, $l_{\max} = 1$, $P = 3$, $M + 1 = 16$ pilots, $n_{\mathrm{iter}} = 2$, $\mathrm{SNR}_p = 50$ dB) on a 1 000-frame Monte-Carlo budget. The reproduced 16-pilot BER curve falls within $\pm 0.2$ log-decade of the visual readout of Zheng's Fig. 3 at $4/6$ SNR cells, with the deviation attributable to the smaller frame budget than Zheng's $10^6$-frame baseline.

A direct head-to-head BER comparison with the present receiver at $N \geq 1024$ is precluded by Zheng's pilot-feasibility constraint $M(Q{+}1) < N{-}Q$, which structurally conflicts with the present operating point of $Q \approx N/3$. The qualitative axis-by-axis structural comparison of Table IX therefore remains the operational comparison; an equal-budget reproduction at a downscaled operating point is flagged as immediate follow-up (§VI-E).

## Appendix D
## Simulator Correspondence and Pilot Convention

All Monte-Carlo runs reported in Section V use an accelerated implementation validated against an independent reference simulator. The implementations produce identical detected QPSK symbols on the $N = 512$ pilot–data realisations used for simulator correspondence; this is why the accelerated implementation is used for the large Monte-Carlo sweeps.

All three implementations use the same superimposed-pilot frame

$$\mathbf{x} = \sqrt{\rho}\,\mathbf{x}_{\mathrm{p}} + \sqrt{1-\rho}\,\mathbf{x}_{\mathrm{d}},$$

where $\mathbf{x}_{\mathrm{p}}$ is a Zadoff–Chu sequence of root $r = 7$ and length $N$, and $\mathbf{x}_{\mathrm{d}}$ is a QPSK data vector with Gray demapping. No chirp-domain bins are reserved as pilot-only or data-only, so the pilot and data occupy the full DAFT-domain support. Table I in Section V lists the canonical physical and algorithmic parameters used across the experiments. The main Monte-Carlo results operate at the oversampled $f_s = 4B$ operating point; the Nyquist $f_s = 2B$ setting is used only for the sensitivity comparison of §V-D.

## Appendix E
## Paired-Bootstrap and Clopper–Pearson Confidence Intervals

Two complementary confidence-interval procedures are used in Section V. Paired-bootstrap CIs are reported alongside the paired BER-reduction figures in the main 4 B BER waterfall because those reductions are paired differences $\delta_i = e_i^{(0)} - e_i^{(1)}$ on the same $(\mathrm{channel}, \mathrm{data}, \mathrm{noise})$ realisation; a non-paired CI would be substantially wider and would mask the contribution of the learned head. Clopper–Pearson exact-binomial CIs are used on the high-SNR per-arm BER numbers because the observed bit-error counts are close to the finite-sample censoring floor, where the binomial distribution is highly skewed and a Wald-type CI would under-cover.

*a) Paired bootstrap (Experiment 1, 3-seed pooled).:* Let $n$ denote the number of evaluated frames at a given SNR, and for frame $i \in \{1, \ldots, n\}$ let $e_i^{(0)}$ and $e_i^{(1)}$ be the bit-error counts of the baseline and the candidate receiver on the same $(\mathrm{channel}, \mathrm{data}, \mathrm{noise})$ realisation. Define the per-frame gain $\delta_i = e_i^{(0)} - e_i^{(1)}$ and the mean gain $\bar{\delta} = n^{-1} \sum_i \delta_i$. The

95% paired-bootstrap CI is computed by (i) drawing $B_{\text{boot}} = 1\,000$ bootstrap resamples of $\{\delta_1, \ldots, \delta_n\}$ with replacement, (ii) computing the resample mean $\bar{\delta}^{(b)}$ for each, and (iii) reporting the 2.5% and 97.5% quantiles of the resample distribution as the CI.

*b) Clopper–Pearson (Experiment 1 headline at high SNR).:* Let $k$ be the observed bit-error count and $n$ the total tested bit count. For $1 \le k \le n-1$, the two-sided Clopper–Pearson 95% CI for the unknown BER $p$ is

$$\big[\, F^{-1}_{k,\,n-k+1}(0.025),\ F^{-1}_{k+1,\,n-k}(0.975)\,\big],$$

where $F^{-1}_{a,b}$ is the inverse CDF of a Beta distribution with parameters $(a, b)$ (equivalent to the exact binomial tail inversion). For the boundary cases the standard Clopper–Pearson convention applies: at $k = 0$ the lower bound is defined to be 0 (the $\mathrm{Beta}(0, n+1)$ distribution is improper) and the upper bound is $1 - 0.025^{1/n}$; at $k = n$ the lower bound is $0.025^{1/n}$ and the upper bound is 1. For the $\mathrm{SNR} = 25$ dB 4 B observation of the learned arm reported in Section V ($k = 4$ bit errors at the adaptive-stop budget of $n \approx 7.17 \times 10^5$ bits), the interior formula yields $\mathrm{CI}_{95\%} \approx [1.52 \times 10^{-6},\ 1.43 \times 10^{-5}]$. For the 4 B PCSI+MMSE reference at $\mathrm{SNR} = 25$ dB, the oracle arm runs to the full nominal budget $n = 8.19 \times 10^5$ bits with $k = 0$, so the upper CP bound is $1 - 0.025^{1/n} \approx 4.5 \times 10^{-6}$. This is what we mean by "finite-sample censoring" in Sections V and VI.

## Appendix F
## OMP-Once Recall Against the Oracle Dirichlet Support (Probe (a))

Probe (a) of §V-E is computed directly from the per-candidate binary labels produced during head training, with no fresh Monte-Carlo required.

*Support-set definitions (crucial; depends on §III-H and (2)).* Three distinct support sets are involved in this probe and must be kept separate because the recall bookkeeping depends on which of the three is on which side of the intersection.

- $S^{\text{phys}}_{\text{OMP}}$ is the *physical* top-$P$ integer $(l, k)$-bin support returned by a single OMP-once pass run with target sparsity equal to the physical path count: $|S^{\text{phys}}_{\text{OMP}}| = P = 8$.
- $\tilde{S}_{\text{OMP}}$ is the *Dirichlet-expansion* of $S^{\text{phys}}_{\text{OMP}}$ into the virtual-tap lattice obtained by replicating each physical bin into the $2k_{\text{leak}} + 1$ adjacent Doppler bins per (2); nominally $|\tilde{S}_{\text{OMP}}| = P \cdot (2k_{\text{leak}} + 1) = 40$.
- $S_{\text{oracle}}$ is the *Dirichlet-weighted virtual-tap* oracle support, i.e. the set of virtual taps that the simulator's stored channel parameters and (2) actually populate. Nominally $|S_{\text{oracle}}| \le P \cdot (2k_{\text{leak}}+1) = 40$; because the per-tap Dirichlet coefficient (2) is small for fractional residuals near zero, only an *effective* mean of $p_{\text{real}} \approx 28$ virtual taps per frame are non-negligible, and the probe stores this per-frame effective count.

Note that $|S^{\text{phys}}_{\text{OMP}}| = P = 8$ and $|\tilde{S}_{\text{OMP}}| \le 40$ generally differ from $|S_{\text{oracle}}| \approx 28$; consequently, recall and precision are distinct quantities for this probe.

For each evaluated frame, the label-generation pass runs the physical OMP-once receiver to obtain $S^{\text{phys}}_{\text{OMP}}$, computes both $\tilde{S}_{\text{OMP}}$ and $S_{\text{oracle}}$, and assigns $y_i = 1$ iff candidate $i$ is an oracle virtual tap that is not covered by the Dirichlet-expanded OMP support. The evaluation record also stores $p_{\text{real}} = |S_{\text{oracle}}|$ per frame.

Recall against the oracle virtual-tap support is

$$\text{recall} \;=\; \frac{|\tilde{S}_{\text{OMP}} \cap S_{\text{oracle}}|}{|S_{\text{oracle}}|} \;=\; \frac{p_{\text{real}} - \sum_i y_i}{p_{\text{real}}}.$$

A consistency cross-check on the aggregates is that the table-level mean recall $\overline{\text{recall}} \approx 0.953$ and mean missed-tap count $\overline{\sum_i y_i} \approx 1.31$ imply $1.31/(1 - 0.953) \approx 28$, matching the stored mean $\overline{p_{\text{real}}}$ to within rounding.

Precision, by contrast, equals $|\tilde{S}_{\text{OMP}} \cap S_{\text{oracle}}|/|\tilde{S}_{\text{OMP}}|$ and is in general different from recall because $|\tilde{S}_{\text{OMP}}| \ne |S_{\text{oracle}}|$; Table IV reports recall only and does not rely on a precision-equal-to-recall claim.

Per-frame and per-SNR aggregates are computed from this formula directly from the stored labels.

## Appendix G
## McNemar Paired-Bit Significance Test

On a paired bit-level comparison between the classical joint and the learned receiver, let $n_{10}$ denote the number of bits on which the classical is wrong and the learned is correct, and $n_{01}$ the number of bits on which the classical is correct and the learned is wrong. The continuity-corrected McNemar $\chi^2$ statistic is

$$\chi^2_{\text{MN}} \;=\; \frac{(|n_{10} - n_{01}| - 1)^2}{n_{10} + n_{01}}, \qquad (n_{10} + n_{01} \ge 1)$$

which is approximately $\chi^2_1$-distributed under the null hypothesis that the two receivers have the same bit-wise error probability and that the discordant-pair indicators are mutually independent. The *two-sided asymptotic* $p$-value $p_{\text{two}} = 1 - F_{\chi^2_1}(\chi^2_{\text{MN}})$ is the value reported in Table II; this is the standard continuity-corrected McNemar convention. For the degenerate case $n_{10} + n_{01} = 0$ (no discordant pairs at all) the statistic is undefined; we report this as "not informative ($p = 1$)" and exclude that SNR cell from the significance summary. For very small discordant counts (rule-of-thumb $n_{10} + n_{01} < 25$), the asymptotic $\chi^2_1$ approximation is unreliable, and we instead use the exact one-sided binomial sign test described next. When a strictly one-sided statement is required ("learned better"), we supplement it with the exact one-sided binomial sign test on the discordant pairs: $p_{\text{one-sided}} = \Pr[\mathrm{Bin}(n_{10} + n_{01}, 1/2) \ge n_{10}]$, computed from the exact binomial tail; because $n_{10}$ dominates $n_{01}$ by multiple orders of magnitude at every high-SNR point where the test is applied, the exact one-sided value tracks the asymptotic two-sided value to within a numerical rounding of each other, and the qualitative conclusion ("learned strictly better at every SNR where the test is applicable") is invariant to the choice of side.

*a) Independence assumption: a known caveat.:* The McNemar test as classically derived assumes that each discordant indicator is an independent Bernoulli draw under $H_0$. In our application the unit of analysis is the *bit*, and bits within a single $(\text{channel}, \text{data}, \text{noise})$ frame share a common $H_{\text{eff}}$ realisation; within-frame discordant indicators are therefore positively correlated (a frame in which the learned receiver

helps tends to help on most of its bits at once). This violates the strict independence assumption, and the consequence is that the effective number of independent Bernoulli draws underlying the test is less than $n_{10} + n_{01}$. The resulting $p$-values are anti-conservative relative to a frame-clustered alternative such as a frame-level paired sign test or a frame-clustered bootstrap. We therefore quote the bit-level McNemar $p$-values as a within-protocol consistency indicator rather than as the output of an independence-respecting exact test, and the qualitative result on which we rely — consistent paired BER reduction across three independent evaluation seeds (Table II) — remains the binding observation.

## APPENDIX H
## ANALYTICAL DECOMPOSITION OF THE FIXED-BUDGET TOP-$K$ DEPLOYMENT RULE

This appendix expands the ranking-only deployment argument summarised in §IV. It is included because the deployed inference rule uses a fixed-budget top-$K_{\text{add}}$ selector rather than a calibrated probability threshold.

Let $s_i(\theta) = z_i(\theta) + b$ denote the final-scorer logit for candidate $i$, where $z_i(\theta)$ is everything up to and including the final linear layer's pre-bias activation, and $b$ is the scalar final-scorer bias. Replacing the learned bias $b_{\text{learned}} \approx -8$ by a deployment bias $b_{\text{deploy}} = +10$ is equivalent to the uniform translation $s_i \mapsto s_i + \Delta b$ with $\Delta b = b_{\text{deploy}} - b_{\text{learned}} = +18$, applied to every candidate's logit. The deployed selection is the composition of two operations: first a per-candidate *gate* $g_i := \mathbf{1}\{\sigma(s_i) > 0.5\} = \mathbf{1}\{s_i > 0\}$, then a top-$K_{\text{add}}$ ranker over the gated set. The two operations decompose cleanly:

1) **Rank preservation**: conditional on the same set of candidates passing the gate, the top-$K_{\text{add}}$ output is invariant under uniform translation, because $\operatorname{argtop}_K(s_1 + \Delta b, \ldots, s_M + \Delta b) = \operatorname{argtop}_K(s_1, \ldots, s_M)$ for any $\Delta b \in \mathbb{R}$, assuming the same deterministic tie-breaking rule is applied to both inputs.
2) **Gate shift**: the deployment bias shifts the gate threshold from "$z_i > 8$" (typical for $b_{\text{learned}} = -8$, almost never satisfied) to "$z_i > -10$" (typical for $b_{\text{deploy}} = +10$, almost always satisfied), so every frame proceeds to a fixed-budget top-$K_{\text{add}}$ selection rather than to a no-op gate output.

The combined behaviour converts a thresholded selector into a frame-unconditional fixed-budget top-$K_{\text{add}}$ ranker. This is intentional: the head is reported as a deterministic fixed-budget ranker, not as a calibrated detector. The deployment bias does not alter the relative ordering of the selected atoms; the learned logits are consumed downstream only through $\operatorname{argtop}_{K_{\text{add}}}$, and probability calibration of the scores is out of scope.

## APPENDIX I
## OMP DICTIONARY ATOM CONSTRUCTION AND ALGORITHM STEPS

This appendix gives an explicit construction of the chirp-domain dictionary atoms used by the OMP step of Algorithm 1 and writes out the OMP iteration with per-step complexity annotations, recovering the bound $C_{\text{OMP}} = \mathcal{O}(PN^2)$.

### *F.1 Dictionary atom construction*

The OMP dictionary $\mathbf{\Phi} \in \mathbb{C}^{N \times Q}$ has one column per integer delay–Doppler bin $(l, k)$ with $l \in \{0, \ldots, l_{\max} - 1\}$ and $k \in \{-k_{\max}, \ldots, k_{\max}\}$, so $Q = (2k_{\max}+1)\, l_{\max}$. The $(l, k)$ atom is the chirp-domain response that the unit-energy Zadoff–Chu pilot $\mathbf{x}_{\text{p}}$ would produce if the channel consisted of a single path of unit gain with integer delay $l$ and integer Doppler $k$. Writing the IDAFT of (1) in matrix form as $\mathbf{F}^{-1}$ and the forward DAFT as $\mathbf{F}$, the time-domain cyclic delay operator as $\mathbf{T}_l$ (the cyclic shift by $l$ samples), and the time-domain Doppler shift as $\mathbf{D}_k = \operatorname{diag}\big(e^{j2\pi kn/N}\big)_{n=0}^{N-1}$, the atom is

$$\boldsymbol{\phi}_{(l,k)} \;=\; \sqrt{\rho}\, \mathbf{F}\, \mathbf{D}_k\, \mathbf{T}_l\, \mathbf{F}^{-1}\, \mathbf{x}_{\text{p}} \;\in\; \mathbb{C}^N, \tag{9}$$

which encodes the round trip "transmit pilot through the chirp domain $\rightarrow$ apply delay-$l$ Doppler-$k$ in time $\rightarrow$ observe in the chirp domain". Because $\mathbf{F}$ is unitary, $\mathbf{T}_l$ is a permutation matrix, and $\mathbf{D}_k$ has unit-modulus diagonal entries, the energy of every atom is exactly the pilot fraction: $\|\boldsymbol{\phi}_{(l,k)}\|^2 = \rho \cdot \|\mathbf{x}_{\text{p}}\|^2 = \rho$ for all $(l, k)$. This places OMP's matched-filter scoring on a common energy scale across the dictionary and is the reason no per-column normalisation is needed inside the OMP iteration.

The dictionary spans physical integer $(l, k)$ bins only. Each fractional-Doppler physical path $(\tau_p, k_p^{\text{real}})$ of §III-B excites one main-lobe column $\boldsymbol{\phi}_{(l_p,k_p)}$ together with $2k_{\text{leak}}$ side-lobe columns at $(l_p, k_p \pm 1), \ldots, (l_p, k_p \pm k_{\text{leak}})$ through the Dirichlet kernel of (2); OMP recovers a subset of these columns directly, and the post-OMP DA-LS refit and MP-BP detection blocks of §III-E operate on the $(2k_{\text{leak}}+1)$-fold expanded virtual-tap representation. The fractional-Doppler half-width $k_{\text{leak}}$ therefore does not enter the dictionary size $Q$; it enters only the downstream blocks via the Dirichlet expansion.

### *F.2 OMP algorithm steps*

Given the chirp-domain observation $\mathbf{y}$ and the dictionary $\mathbf{\Phi}$, OMP recovers a sparse support $\hat{S}$ of size $P$ together with the corresponding least-squares path-gain estimate $\hat{\mathbf{h}}$. The iteration is summarised in Algorithm 2 with per-step complex-multiplication counts.

The dominant per-iteration cost is the matched-filter sweep at $\mathcal{O}(NQ)$ in step 2(a); the least-squares solve at $\mathcal{O}(Ns + s^3)$ is sub-dominant because the support size $s \le P = 8$ remains small throughout and $s^3$ is therefore bounded by a constant much less than $N$. Summing the matched-filter sweep across $P$ iterations and the least-squares cost across $s = 1, \ldots, P$ gives a total of

$$\begin{aligned} C_{\text{OMP}} &= \mathcal{O}(PNQ) + \mathcal{O}\left( N \sum_{s=1}^{P} s + \sum_{s=1}^{P} s^3 \right) \\ &= \mathcal{O}(PNQ + P^2 N + P^4) = \mathcal{O}(PNQ). \end{aligned}$$

Since $P \ll N$ and $Q = \mathcal{O}(N)$, this reduces to $\mathcal{O}(PN^2)$. Thus the matched-filter sweep in step 2(a) is the dominant per-frame block, consistent with the 14–18.6 M MAC estimates at $N = 1024$ in Table VI.

**Algorithm 2** Orthogonal Matching Pursuit on the chirp-domain pilot dictionary [46].

**Input**: chirp-domain observation $\mathbf{y} \in \mathbb{C}^N$, dictionary $\mathbf{\Phi} \in \mathbb{C}^{N\times Q}$, target sparsity $P$.
**Output**: support $\hat{S} \subseteq \{0,\dots,Q-1\}$ with $|\hat{S}| = P$, path-gain vector $\hat{\mathbf{h}} \in \mathbb{C}^P$.

1) Initialise residual $\mathbf{r}^{(0)} = \mathbf{y}$ and support $\hat{S}^{(0)} = \emptyset$.
2) **For** $s = 1,\dots,P$:
   a) Compute the matched-filter response: $\mathbf{c}^{(s)} = \mathbf{\Phi}^H \mathbf{r}^{(s-1)} \in \mathbb{C}^Q$. $(\mathcal{O}(NQ))$
   b) Pick the strongest unselected atom: $\hat{j}^{(s)} = \arg\max_{j\notin\hat{S}^{(s-1)}} |c_j^{(s)}|$. $(\mathcal{O}(Q))$
   c) Append it to the support: $\hat{S}^{(s)} = \hat{S}^{(s-1)} \cup \{\hat{j}^{(s)}\}$.
   d) Solve the restricted least-squares problem on the current support, $\hat{\mathbf{h}}^{(s)} = \mathbf{\Phi}^{\dagger}_{\hat{S}^{(s)}}\,\mathbf{y}$, by QR or normal-equations factorisation. $(\mathcal{O}(Ns + s^3))$
   e) Update the residual: $\mathbf{r}^{(s)} = \mathbf{y} - \mathbf{\Phi}_{\hat{S}^{(s)}}\,\hat{\mathbf{h}}^{(s)}$. $(\mathcal{O}(Ns))$
3) Return $\hat{S} = \hat{S}^{(P)}$ and $\hat{\mathbf{h}} = \hat{\mathbf{h}}^{(P)}$.

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